Supplementary Materials for

Crop water-management priorities depend on the accounting metric: A remote-sensing assessment across Pakistan's Indus Basin

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This file includes Sections S1-S12, Tables S1-S12, and Figures S1-S17. Complete machine-readable versions of all tables, analytical rasters, boundary files, and code are archived at https://github.com/Ahmad5046/indus-basin-crop-water-supplementary-materials. The immutable repository revision used for submission is identified in the manuscript's Data Availability and Supplementary Materials statements.

S1. Analytical scope and reproducibility

The analytical support is cropland within the Pakistan portion of the Indus drainage basin. It is not an official canal-command delineation. The crop map describes the 2023 seasonal extent: crop-hectares can exceed unique physical area under double cropping, maize combines spring and autumn cycles, and sugarcane is treated as annual. Water accounting is crop-season based and residual blue ET is not measured irrigation withdrawal.

S2. Predictor and classification details

The Rabi stack contains 42 predictors and the Kharif stack 47. Predictor families include period-specific Sentinel-2 NDVI, EVI, NDWI, LSWI, and SAVI; full-season Sentinel-2 and Landsat optical bands and indices; Sentinel-1 median VV, VH, and VV-VH; and SRTM elevation and slope. Missing predictors were filled with -0.25 before sampling. Both models used 300 trees, minimum leaf population 2, bag fraction 0.70, default variables per split, no maximum-node limit, and classifier seed 20230709. Reference points were assigned by whole 0.05-degree blocks using a deterministic SHA-256 rule with seed 20230709.

Table S1. Class-level validation diagnostics

Season	Class Code	Crop Name	Validation Support	Predicted Support	Correct	Producer Accuracy	User Accuracy	F1
Rabi	1	Wheat	467.000	549.000	438.000	0.938	0.798	0.862
Rabi	4	Sugarcane	125.000	115.000	97.000	0.776	0.843	0.808
Rabi	5	Maize	214.000	203.000	193.000	0.902	0.951	0.926
Rabi	6	Other cropland	474.000	413.000	365.000	0.770	0.884	0.823
Kharif	2	Rice	517.000	523.000	478.000	0.925	0.914	0.919
Kharif	3	Cotton	320.000	339.000	284.000	0.887	0.838	0.862
Kharif	4	Sugarcane	436.000	454.000	397.000	0.911	0.874	0.892
Kharif	5	Maize	587.000	614.000	491.000	0.836	0.800	0.818
Kharif	6	Other cropland	529.000	459.000	380.000	0.718	0.828	0.769

Table S2. Season-level validation diagnostics

Season	Overall Accuracy	Kappa	Balanced Accuracy	Macro F1	Training Samples	Validation Samples	Validation Blocks	Interpretation
Rabi	0.854	0.788	0.846	0.855	5535.000	1280.000	NA	spatial-block holdout; not area-probability sampled

Season	Overall Accuracy	Kappa	Balanced Accuracy	Macro F1	Training Samples	Validation Samples	Validation Blocks	Interpretation
Kharif	0.850	0.811	0.855	0.852	9763.000	2389.000	NA	spatial-block holdout; not area-probability sampled

S3. Crop-area scenario

The official-area-constrained scenario multiplies mapped mean crop water depth and mapped yield composition by the ratio of official harvested area to mapped crop-season area. It is a first-order structural scenario, not a confidence interval. Because area and production are scaled by the same factor, crop-level VWF remains unchanged while water volume and production scale directly.

Table S3. Mapped and official-area-constrained crop scenarios

Crop Name	Mapped Area Ha	Official Harvested Area Ha	Mapped Minus Official Pct	Mapped Total Water Km3	Official Area Scaled Total Water Km3	Mapped Total Vwf M3 T
Wheat	7,930,296	8,477,154	-6.451	26.468	28.293	1066.851
Rice	4,156,694	2,969,151	39.996	15.284	10.918	1548.604
Cotton	1,293,009	1,978,787	-34.656	5.662	8.665	2076.197
Sugarcane	1,647,041	1,284,439	28.230	12.965	10.110	125.629
Maize	2,770,262	1,710,539	61.953	19.635	12.124	1237.948

S4. Product-chain structural sensitivity

The primary endpoint combines WaPOR v3 AETI with CHIRPS rainfall; the comparison endpoint combines MODIS ET with ERA5-Land rainfall. Endpoints are reported directly. No uniform interpolation, Monte Carlo distribution, or confidence interval is used.

Table S4. Product-chain endpoints

Crop Name	Product Chain	Total Water Mm	Green Water Mm	Blue Water Mm	Total Vwf M3 T	Water Productivity Kg M3
Wheat	WaPOR-CHIRPS	333.753	86.571	247.182	1066.851	0.937
Wheat	MODIS-ERA5-Land	133.591	86.845	46.746	427.028	2.342
Rice	WaPOR-CHIRPS	367.707	225.058	142.649	1548.604	0.646
Rice	MODIS-ERA5-Land	98.240	97.696	0.544	413.738	2.417
Cotton	WaPOR-CHIRPS	437.882	197.116	240.766	2076.197	0.482
Cotton	MODIS-ERA5-Land	116.879	113.785	3.094	554.177	1.804
Sugarcane	WaPOR-CHIRPS	787.152	321.902	465.251	125.629	7.960
Sugarcane	MODIS-ERA5-Land	241.894	226.641	15.253	38.606	25.903
Maize	WaPOR-CHIRPS	708.773	371.337	337.436	1237.948	0.808
Maize	MODIS-ERA5-Land	258.035	255.168	2.867	450.686	2.219

Table S5. Direct endpoint structural envelope

Crop Name	Metric	Primary Wapor Chirps	Comparison Modis Era5 Land	Absolute Difference	Comparison Minus Primary Pct

Crop Name	Metric	Primary Wapor Chirps	Comparison Modis Era5 Land	Absolute Difference	Comparison Minus Primary Pct
Wheat	green_water_mm	86.571	86.845	0.274	0.317
Wheat	blue_water_mm	247.182	46.746	-200.436	-81.088
Wheat	total_water_mm	333.753	133.591	-200.162	-59.973
Wheat	green_vwf_m3_t	276.726	277.603	0.877	0.317
Wheat	blue_vwf_m3_t	790.125	149.426	-640.699	-81.088
Wheat	total_vwf_m3_t	1066.851	427.028	-639.823	-59.973
Wheat	water_productivity_kg_m3	0.937	2.342	1.404	149.831
Wheat	blue_dependency_pct	74.061	34.992	-39.069	-52.753
Rice	green_water_mm	225.058	97.696	-127.362	-56.591
Rice	blue_water_mm	142.649	0.544	-142.105	-99.619
Rice	total_water_mm	367.707	98.240	-269.467	-73.283
Rice	green_vwf_m3_t	947.836	411.448	-536.387	-56.591
Rice	blue_vwf_m3_t	600.768	2.289	-598.479	-99.619
Rice	total_vwf_m3_t	1548.604	413.738	-1134.866	-73.283
Rice	water_productivity_kg_m3	0.646	2.417	1.771	274.296
Rice	blue_dependency_pct	38.794	0.553	-38.241	-98.574
Cotton	green_water_mm	197.116	113.785	-83.331	-42.275
Cotton	blue_water_mm	240.766	3.094	-237.672	-98.715
Cotton	total_water_mm	437.882	116.879	-321.003	-73.308
Cotton	green_vwf_m3_t	934.614	539.506	-395.108	-42.275
Cotton	blue_vwf_m3_t	1141.583	14.670	-1126.912	-98.715
Cotton	total_vwf_m3_t	2076.197	554.177	-1522.020	-73.308
Cotton	water_productivity_kg_m3	0.482	1.804	1.323	274.645
Cotton	blue_dependency_pct	54.984	2.647	-52.337	-95.185
Sugarcane	green_water_mm	321.902	226.641	-95.260	-29.593
Sugarcane	blue_water_mm	465.251	15.253	-449.998	-96.722
Sugarcane	total_water_mm	787.152	241.894	-545.259	-69.270
Sugarcane	green_vwf_m3_t	51.375	36.172	-15.203	-29.593
Sugarcane	blue_vwf_m3_t	74.254	2.434	-71.819	-96.722
Sugarcane	total_vwf_m3_t	125.629	38.606	-87.023	-69.270
Sugarcane	water_productivity_kg_m3	7.960	25.903	17.943	225.412
Sugarcane	blue_dependency_pct	59.106	6.305	-52.800	-89.332
Maize	green_water_mm	371.337	255.168	-116.169	-31.284
Maize	blue_water_mm	337.436	2.867	-334.569	-99.150
Maize	total_water_mm	708.773	258.035	-450.738	-63.594
Maize	green_vwf_m3_t	648.580	445.678	-202.902	-31.284
Maize	blue_vwf_m3_t	589.367	5.007	-584.360	-99.150
Maize	total_vwf_m3_t	1237.948	450.686	-787.262	-63.594
Maize	water_productivity_kg_m3	0.808	2.219	1.411	174.681
Maize	blue_dependency_pct	47.608	1.111	-46.497	-97.666

S5. Fixed-extent temporal context

The 2018-2022 baseline and 2023 values use the fixed 2023 crop masks. They describe hydroclimatic/product variation over a short context window, not historical crop-area change or a long-term climatology.

Table S6. Fixed-2023-extent temporal context

Crop Name	Metric	Baseline Years	Baseline N	Baseline Mean	Baseline Sd	Value 2023	Departure	Departure Pct	Standardized Departure	Scope Note

Crop Name	Metric	Baseline Years	Baseline N	Baseline Mean	Baseline Sd	Value 2023	Departure	Departure Pct	Standardized Departure	Scope Note
Wheat	WaPOR AETI	2018-2022	5.000	282.907	23.375	305.578	22.670	8.013	0.970	fixed validated 2023 crop extent; short product context, not long-term climatology
Wheat	CHIRPS rainfall	2018-2022	5.000	171.499	57.682	115.911	-55.589	-32.413	-0.964	fixed validated 2023 crop extent; short product context, not long-term climatology
Wheat	Primary blue dependency	2018-2022	5.000	48.902	11.581	59.804	10.903	22.295	0.941	fixed validated 2023 crop extent; short product context, not long-term climatology
Rice	WaPOR AETI	2018-2022	5.000	447.013	18.216	385.396	-61.617	-13.784	-3.383	fixed validated 2023 crop extent; short product context, not long-term climatology
Rice	CHIRPS rainfall	2018-2022	5.000	307.427	47.088	318.187	10.760	3.500	0.229	fixed validated 2023 crop extent; short product context, not long-term climatology
Rice	Primary blue dependency	2018-2022	5.000	47.872	7.600	39.174	-8.698	-18.169	-1.144	fixed validated 2023 crop extent; short product context, not long-term climatology
Cotton	WaPOR AETI	2018-2022	5.000	501.249	38.568	408.743	-92.506	-18.455	-2.398	fixed validated 2023 crop extent; short product context, not long-term climatology

Crop Name	Metric	Baseline Years	Baseline N	Baseline Mean	Baseline Sd	Value 2023	Departure	Departure Pct	Standardized Departure	Scope Note
Cotton	CHIRPS rainfall	2018-2022	5.000	197.525	86.609	202.559	5.034	2.549	0.058	fixed validated 2023 crop extent; short product context, not long-term climatology
Cotton	Primary blue dependency	2018-2022	5.000	62.038	9.703	51.626	-10.412	-16.783	-1.073	fixed validated 2023 crop extent; short product context, not long-term climatology
Sugarcane	WaPOR AETI	2018-2022	5.000	876.973	34.337	798.408	-78.564	-8.959	-2.288	fixed validated 2023 crop extent; short product context, not long-term climatology
Sugarcane	CHIRPS rainfall	2018-2022	5.000	384.319	78.809	359.750	-24.569	-6.393	-0.312	fixed validated 2023 crop extent; short product context, not long-term climatology
Sugarcane	Primary blue dependency	2018-2022	5.000	59.253	7.086	57.298	-1.955	-3.299	-0.276	fixed validated 2023 crop extent; short product context, not long-term climatology
Maize	WaPOR AETI	2018-2022	5.000	713.881	16.826	665.658	-48.223	-6.755	-2.866	fixed validated 2023 crop extent; short product context, not long-term climatology
Maize	CHIRPS rainfall	2018-2022	5.000	642.683	63.916	574.696	-67.986	-10.579	-1.064	fixed validated 2023 crop extent; short product context, not long-term climatology

Crop Name	Metric	Baseline Years	Baseline N	Baseline Mean	Baseline Sd	Value 2023	Departure	Departure Pct	Standardized Departure	Scope Note
Maize	Primary blue dependency	2018-2022	5.000	32.323	4.722	31.807	-0.516	-1.595	-0.109	fixed validated 2023 crop extent; short product context, not long-term climatology

S6. Hotspot threshold sensitivity

The district score adds one point each for high blue-water volume, high total VWF, and low WP. Threshold pairs 70/30, 75/25, 80/20, and 90/10 were applied both to all 108 computable districts and to the 76 districts with at least 25,000 mapped crop-hectares. These are internal diagnostic priorities, not environmental or legal thresholds.

Table S7. Hotspot-count sensitivity

Scope	District Count	Threshold Pair	High Quantile	Low Quantile	Blue Volume Cutoff Km3	Total Vwf Cutoff M3 T	Water Productivity Cutoff Kg M3	Score 0 Count	Score 1 Count	Score 2 Count	Score 3 Count	Priority Score Ge 2 Count
all computable districts	108	70/30	0.700	0.300	0.606	1658.319	0.603	42	33	33	0	33
all computable districts	108	75/25	0.750	0.250	0.713	2012.244	0.497	54	27	27	0	27
all computable districts	108	80/20	0.800	0.200	0.818	2302.369	0.434	64	22	22	0	22
all computable districts	108	90/10	0.900	0.100	1.181	3122.410	0.320	86	11	11	0	11
districts with mapped area >=25,000 ha	76	70/30	0.700	0.300	0.791	988.038	1.015	31	22	22	1	23
districts with mapped area >=25,000 ha	76	75/25	0.750	0.250	0.996	1148.249	0.872	38	19	19	0	19
districts with mapped area >=25,000 ha	76	80/20	0.800	0.200	1.139	1444.592	0.692	44	16	16	0	16
districts with mapped area >=25,000 ha	76	90/10	0.900	0.100	1.281	2451.910	0.408	60	8	8	0	8

Table S8. District hotspot stability (first 40 rows; complete CSV supplied separately)

Scope	Gaul1 Name	Gaul2 Name	Mapped Area Ha	Threshold Tests	Priority Count	Minimum Score	Maximum Score	Priority Stability

Scope	Gaul1 Name	Gaul2 Name	Mapped Area Ha	Threshold Tests	Priority Count	Minimum Score	Maximum Score	Priority Stability
all computable districts	Khyber Pakhtunkhwa	Buner	64,589	4.000	4	2.000	2.000	priority under all thresholds
all computable districts	Islamabad	Islamabad	32,749	4.000	4	2.000	2.000	priority under all thresholds
all computable districts	Khyber Pakhtunkhwa	Khyber	15,704	4.000	4	2.000	2.000	priority under all thresholds
all computable districts	Khyber Pakhtunkhwa	Orakzai	14,550	4.000	4	2.000	2.000	priority under all thresholds
all computable districts	Khyber Pakhtunkhwa	Abbottabad	14,007	4.000	4	2.000	2.000	priority under all thresholds
all computable districts	Khyber Pakhtunkhwa	South Waziristan	11,463	4.000	4	2.000	2.000	priority under all thresholds
all computable districts	Khyber Pakhtunkhwa	Karak	10,357	4.000	4	2.000	2.000	priority under all thresholds
all computable districts	Khyber Pakhtunkhwa	Batagram	6,423	4.000	4	2.000	2.000	priority under all thresholds
all computable districts	Khyber Pakhtunkhwa	Shangla	5,773	4.000	4	2.000	2.000	priority under all thresholds
all computable districts	Khyber Pakhtunkhwa	Tor Ghar	1,762	4.000	4	2.000	2.000	priority under all thresholds
all computable districts	Balochistan	Pishin	154	4.000	4	2.000	2.000	priority under all thresholds
all computable districts	Punjab	Rawalpindi	246,278	4.000	3	0.000	2.000	threshold-sensitive priority
all computable districts	Khyber Pakhtunkhwa	Haripur	66,371	4.000	3	0.000	2.000	threshold-sensitive priority
all computable districts	Khyber Pakhtunkhwa	Bajaur	59,996	4.000	3	0.000	2.000	threshold-sensitive priority
all computable districts	Khyber Pakhtunkhwa	Lower Dir	52,253	4.000	3	0.000	2.000	threshold-sensitive priority
all computable districts	Khyber Pakhtunkhwa	Mansehra	43,831	4.000	3	0.000	2.000	threshold-sensitive priority
all computable districts	Khyber Pakhtunkhwa	Swat	43,208	4.000	3	0.000	2.000	threshold-sensitive priority
all computable districts	Khyber Pakhtunkhwa	Kurram	29,969	4.000	3	0.000	2.000	threshold-sensitive priority
all computable districts	Khyber Pakhtunkhwa	Hangu	24,869	4.000	3	0.000	2.000	threshold-sensitive priority
all computable districts	Balochistan	Kachhi	21,531	4.000	3	0.000	2.000	threshold-sensitive priority
all computable districts	Balochistan	Ziarat	6,593	4.000	3	0.000	2.000	threshold-sensitive priority
all computable districts	Khyber Pakhtunkhwa	Kolai Palas Kohistan	1,072	4.000	3	0.000	2.000	threshold-sensitive priority
all computable districts	Balochistan	Jhal Magsi	40,466	4.000	2	0.000	2.000	threshold-sensitive priority
all computable districts	Khyber Pakhtunkhwa	Mohmand	22,926	4.000	2	0.000	2.000	threshold-sensitive priority
all computable districts	Balochistan	Sibi	20,280	4.000	2	0.000	2.000	threshold-sensitive priority
all computable districts	Khyber Pakhtunkhwa	Upper Dir	13,051	4.000	2	0.000	2.000	threshold-sensitive priority

Scope	Gaul1 Name	Gaul2 Name	Mapped Area Ha	Threshold Tests	Priority Count	Minimum Score	Maximum Score	Priority Stability
all computable districts	Khyber Pakhtunkhwa	North Waziristan	13,000	4.000	2	0.000	2.000	threshold-sensitive priority
all computable districts	Punjab	Chakwal	188,878	4.000	1	0.000	2.000	threshold-sensitive priority
all computable districts	Balochistan	Nasirabad	166,279	4.000	1	0.000	2.000	threshold-sensitive priority
all computable districts	Balochistan	Killa Saifullah	24,526	4.000	1	0.000	2.000	threshold-sensitive priority
all computable districts	Balochistan	Musakhel	9,878	4.000	1	0.000	2.000	threshold-sensitive priority
all computable districts	Balochistan	Mastung	2,891	4.000	1	0.000	2.000	threshold-sensitive priority
all computable districts	Balochistan	Sherani	1,090	4.000	1	0.000	2.000	threshold-sensitive priority
all computable districts	Punjab	Jhang	590,741	4.000	0	1.000	1.000	never priority
all computable districts	Punjab	Muzaffargarh	576,725	4.000	0	1.000	1.000	never priority
all computable districts	Punjab	Okara	552,483	4.000	0	1.000	1.000	never priority
all computable districts	Punjab	Faisalabad	531,342	4.000	0	1.000	1.000	never priority
all computable districts	Punjab	Sargodha	518,491	4.000	0	1.000	1.000	never priority
all computable districts	Punjab	Sheikhupura	510,194	4.000	0	0.000	1.000	never priority
all computable districts	Punjab	Vehari	508,577	4.000	0	1.000	1.000	never priority

S7. Yield coverage

Production-dependent indicators were calculated only for positive matched district yields. No unsupported yield was imputed. Of 580 district-crop combinations, 379 were computable and 201 lacked a positive matched yield or mapped crop area.

Table S9. Yield coverage by crop

Crop Name	District Crop Rows	Rows With Mapped Crop	Rows With Positive Yield	Computable Rows	Mapped Area Ha	Computable Row Pct
Cotton	116.000	116.000	46.000	46.000	1,293,009	39.655
Maize	116.000	115.000	81.000	80.000	2,770,262	68.966
Rice	116.000	114.000	78.000	78.000	4,156,694	67.241
Sugarcane	116.000	115.000	68.000	67.000	1,647,041	57.759
Wheat	116.000	116.000	108.000	108.000	7,930,296	93.103

Table S10. Yield coverage by district (first 50 rows; complete CSV supplied separately)

Gaul1 Name	Gaul2 Name	Crop Rows	Computable Crop Rows	Mapped Area Ha	Any Computable
Balochistan	Barkhan	5.000	3.000	39,240	1.000
Balochistan	Dera Bugti	5.000	4.000	22,754	1.000
Balochistan	Duki	5.000	0.000	16,889	0.000
Balochistan	Harnai	5.000	3.000	3,163	1.000
Balochistan	Jaffarabad	5.000	3.000	137,821	1.000

Gaul1 Name	Gaul2 Name	Crop Rows	Computable Crop Rows	Mapped Area Ha	Any Computable
Balochistan	Jhal Magsi	5.000	2.000	40,466	1.000
Balochistan	Kachhi	5.000	2.000	21,531	1.000
Balochistan	Kalat	5.000	1.000	3,603	1.000
Balochistan	Khuzdar	5.000	3.000	32,972	1.000
Balochistan	Killa Saifullah	5.000	2.000	24,526	1.000
Balochistan	Kohlu	5.000	2.000	11,602	1.000
Balochistan	Lasbela	5.000	2.000	16	1.000
Balochistan	Lehri	5.000	0.000	11,313	0.000
Balochistan	Loralai	5.000	3.000	14,367	1.000
Balochistan	Mastung	5.000	1.000	2,891	1.000
Balochistan	Musakhel	5.000	3.000	9,878	1.000
Balochistan	Nasirabad	5.000	3.000	166,279	1.000
Balochistan	Pishin	5.000	1.000	154	1.000
Balochistan	Quetta	5.000	1.000	27	1.000
Balochistan	Shaheed Sikandarabad	5.000	0.000	2,560	0.000
Balochistan	Sherani	5.000	2.000	1,090	1.000
Balochistan	Sibi	5.000	3.000	20,280	1.000
Balochistan	Sohbatpur	5.000	0.000	69,023	0.000
Balochistan	Zhob	5.000	2.000	7,671	1.000
Balochistan	Ziarat	5.000	2.000	6,593	1.000
Islamabad	Islamabad	5.000	1.000	32,749	1.000
Khyber Pakhtunkhwa	Abbottabad	5.000	2.000	14,007	1.000
Khyber Pakhtunkhwa	Bajaur	5.000	3.000	59,996	1.000
Khyber Pakhtunkhwa	Bannu	5.000	4.000	47,372	1.000
Khyber Pakhtunkhwa	Batagram	5.000	3.000	6,423	1.000
Khyber Pakhtunkhwa	Buner	5.000	4.000	64,589	1.000
Khyber Pakhtunkhwa	Charsadda	5.000	4.000	73,660	1.000
Khyber Pakhtunkhwa	Chitral Lower	5.000	3.000	2,783	1.000
Khyber Pakhtunkhwa	Chitral Upper	5.000	0.000	1,657	0.000
Khyber Pakhtunkhwa	D. I. Khan	5.000	5.000	244,506	1.000
Khyber Pakhtunkhwa	Hangu	5.000	4.000	24,869	1.000
Khyber Pakhtunkhwa	Haripur	5.000	3.000	66,371	1.000
Khyber Pakhtunkhwa	Karak	5.000	2.000	10,357	1.000
Khyber Pakhtunkhwa	Khyber	5.000	3.000	15,704	1.000
Khyber Pakhtunkhwa	Kohat	5.000	4.000	39,270	1.000
Khyber Pakhtunkhwa	Kohistan Lower	5.000	0.000	460	0.000
Khyber Pakhtunkhwa	Kohistan Upper	5.000	0.000	1,331	0.000
Khyber Pakhtunkhwa	Kolai Palas Kohistan	5.000	3.000	1,072	1.000
Khyber Pakhtunkhwa	Kurram	5.000	3.000	29,969	1.000
Khyber Pakhtunkhwa	Lakki Marwat	5.000	4.000	38,615	1.000
Khyber Pakhtunkhwa	Lower Dir	5.000	3.000	52,253	1.000
Khyber Pakhtunkhwa	Malakand	5.000	4.000	25,220	1.000
Khyber Pakhtunkhwa	Mansehra	5.000	3.000	43,831	1.000
Khyber Pakhtunkhwa	Mardan	5.000	4.000	124,589	1.000
Khyber Pakhtunkhwa	Mohmand	5.000	2.000	22,926	1.000

S8. Numerical QC identities

- Green volume + residual-blue volume = total crop water volume within floating-point tolerance.
- Mapped crop area x matched district yield = mapped production.
- Total water volume / mapped production = total VWF.
- Mapped production mass / total water volume = water productivity.
- Unavailable yield remains unavailable rather than being displayed as zero.

S9. Software and execution record

Component	Record
Operating system	Windows-11-10.0.26200-SP0
Python	3.12.13
Word automation	Microsoft Word 16.0 COM
Geospatial processing	Google Earth Engine JavaScript/Python APIs; versions recorded in project environment files where available
Classification/block seed	20230709 (authoritative archived output provenance)
Structural sensitivity seed	Not applicable; direct endpoints only
Primary CRS	EPSG:32642; source geographic products retained in their documented CRSs

S10. Crop calendars

Crop calendars were defined before water aggregation so that crops with different phenology were not represented by a single annual total. Wheat spans the calendar-year boundary, maize combines spring and autumn growing periods, and sugarcane is treated as an annual crop. End dates are exclusive. These windows organize the satellite ET and rainfall accounting and are not substitutes for field-specific crop coefficients.

Table S11. Crop-specific aggregation windows used in classification and water accounting

Crop	Season	Start date	End date (exclusive)	Classification mask
Wheat	Rabi	1 Nov 2022	1 May 2023	Rabi Random Forest map
Rice	Kharif	1 Jun 2023	1 Nov 2023	Kharif Random Forest map
Cotton	Kharif	1 Apr 2023	1 Dec 2023	Kharif Random Forest map
Sugarcane	Annual	1 Jan 2023	1 Jan 2024	Rabi/Kharif union
Maize	Spring/autumn	1 Feb 2023	1 Dec 2023	Rabi/Kharif union

S11. Use of AI-assisted tools during manuscript preparation

Tool: OpenAI Codex.

Period of use: July-October 2026.

Purpose: Language editing, consistency checking among the text, tables, and equations, document-format automation, and assistance in drafting reproducibility and quality-control scripts.

Prompt records: The exact wording of individual prompts was not retained. The examples in Table S12 were reconstructed by the authors to document the type and scope of instructions used in each task category; they are not represented as verbatim records.

Table S12. Representative AI-assisted task categories and verification steps

Task category	Representative instruction (reconstructed)	Verification by the authors
Language editing	"Improve the clarity and grammar of this paragraph without changing its meaning, technical terms, or any numerical values."	Each edit was compared with the original text; changes that altered meaning were rejected.
Consistency checking	"Check whether the values reported in this section match the values in Table X and flag any mismatch."	Each flagged item was checked manually against the derived tables archived in the code repository.
Document formatting	"Format the reference list and section headings according to the journal style."	The formatted document was checked visually.
Reproducibility scripting	"Write a Python script that recomputes the crop-level totals from the exported CSV files and compares them with the manuscript tables."	Scripts were executed and their outputs verified against the Google Earth Engine exports; the scripts are archived in the code repository.

OpenAI Codex was not listed as an author, and its outputs were not treated as scientific evidence. All AI-assisted text and code were reviewed, tested, and, where necessary, revised by the authors. Reported values were checked against the archived Python and Google Earth Engine outputs. The authors take full responsibility for the accuracy, originality, and integrity of the manuscript.

S12. Supplementary figures

Figures S1–S17 are reproduced below at reduced resolution; 1200 dpi PNG versions are archived in the repository. Figure S1 shows the full 116-district blue-VWF ranking, which is retained outside the main manuscript because district labels are not readable at journal panel size.

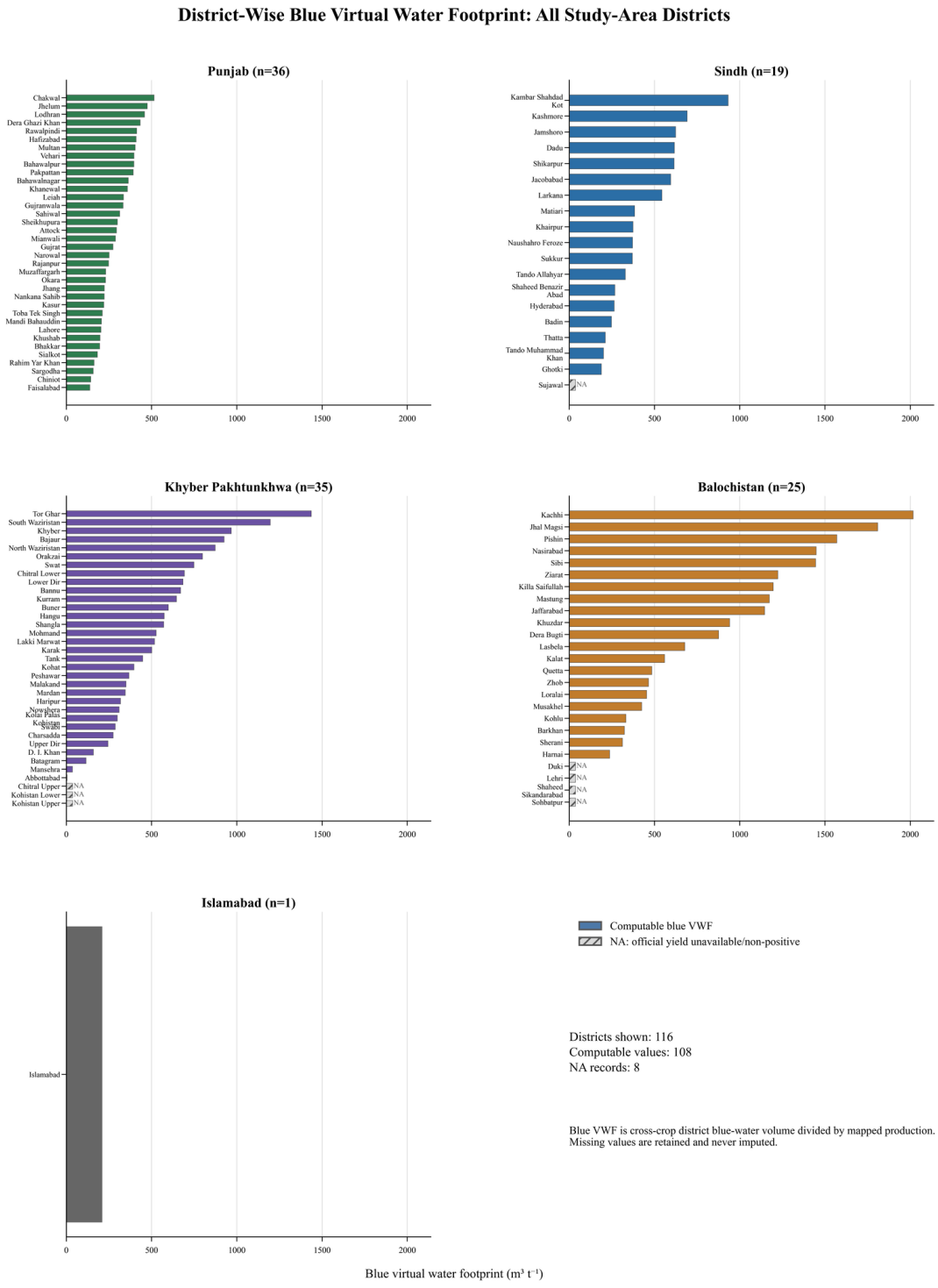


Figure S1. Full ranking of blue virtual water footprint for all 116 study-area district records; non-computable records are retained explicitly.

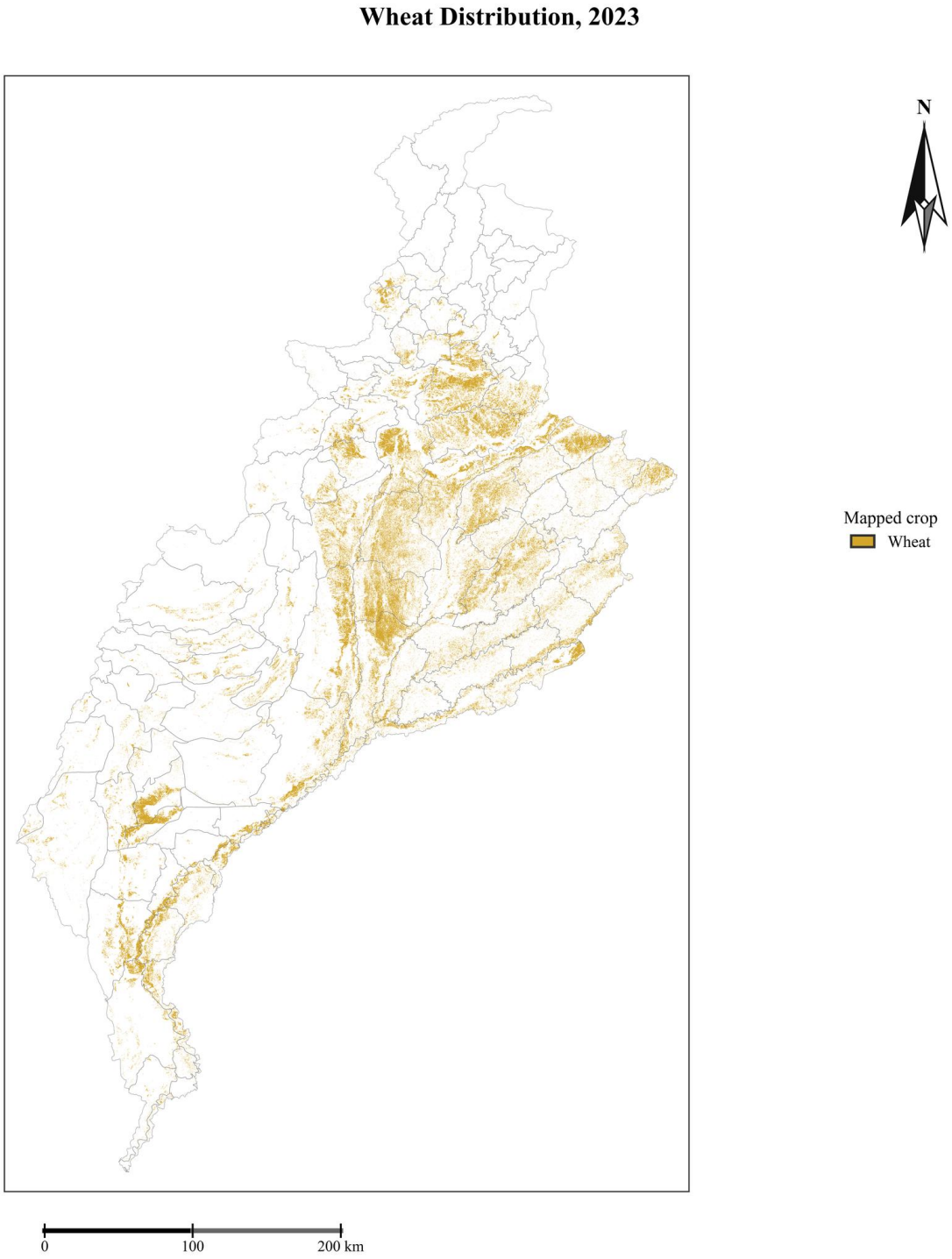


Figure S2. Mapped 2023 wheat distribution within the Pakistan portion of the Indus drainage-basin analysis boundary.

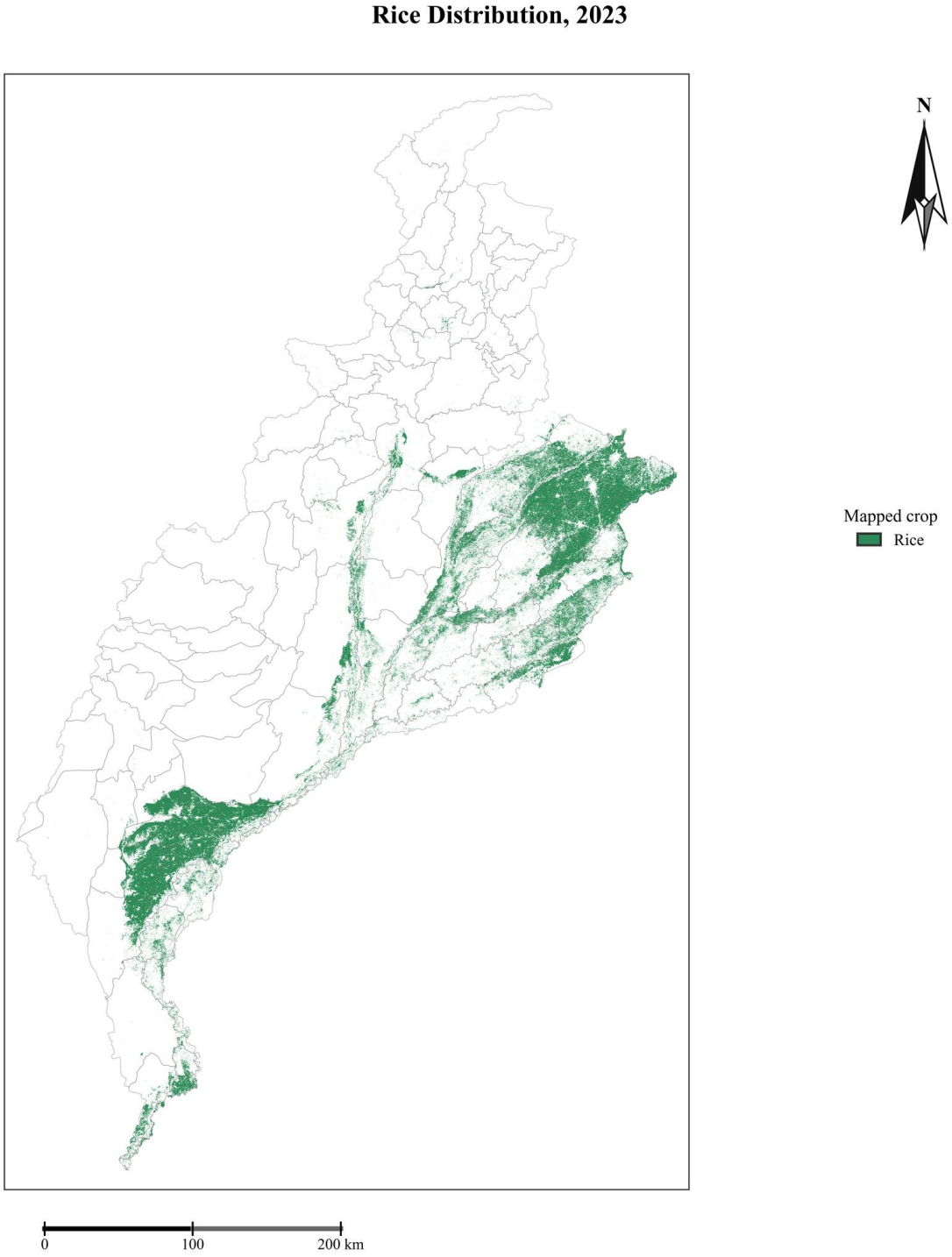


Figure S3. Mapped 2023 rice distribution within the Pakistan portion of the Indus drainage-basin analysis boundary.

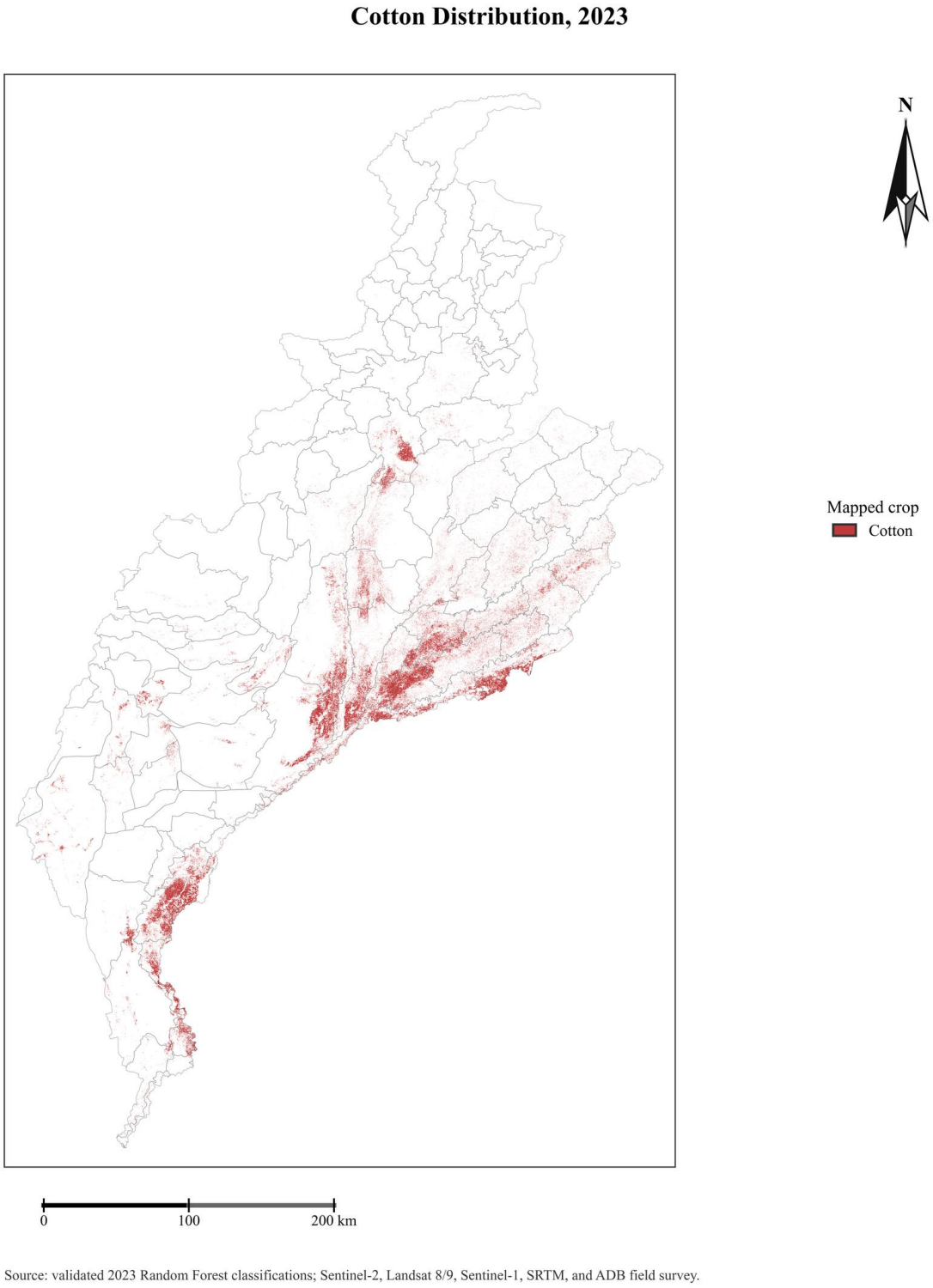


Figure S4. Mapped 2023 cotton distribution within the Pakistan portion of the Indus drainage-basin analysis boundary.

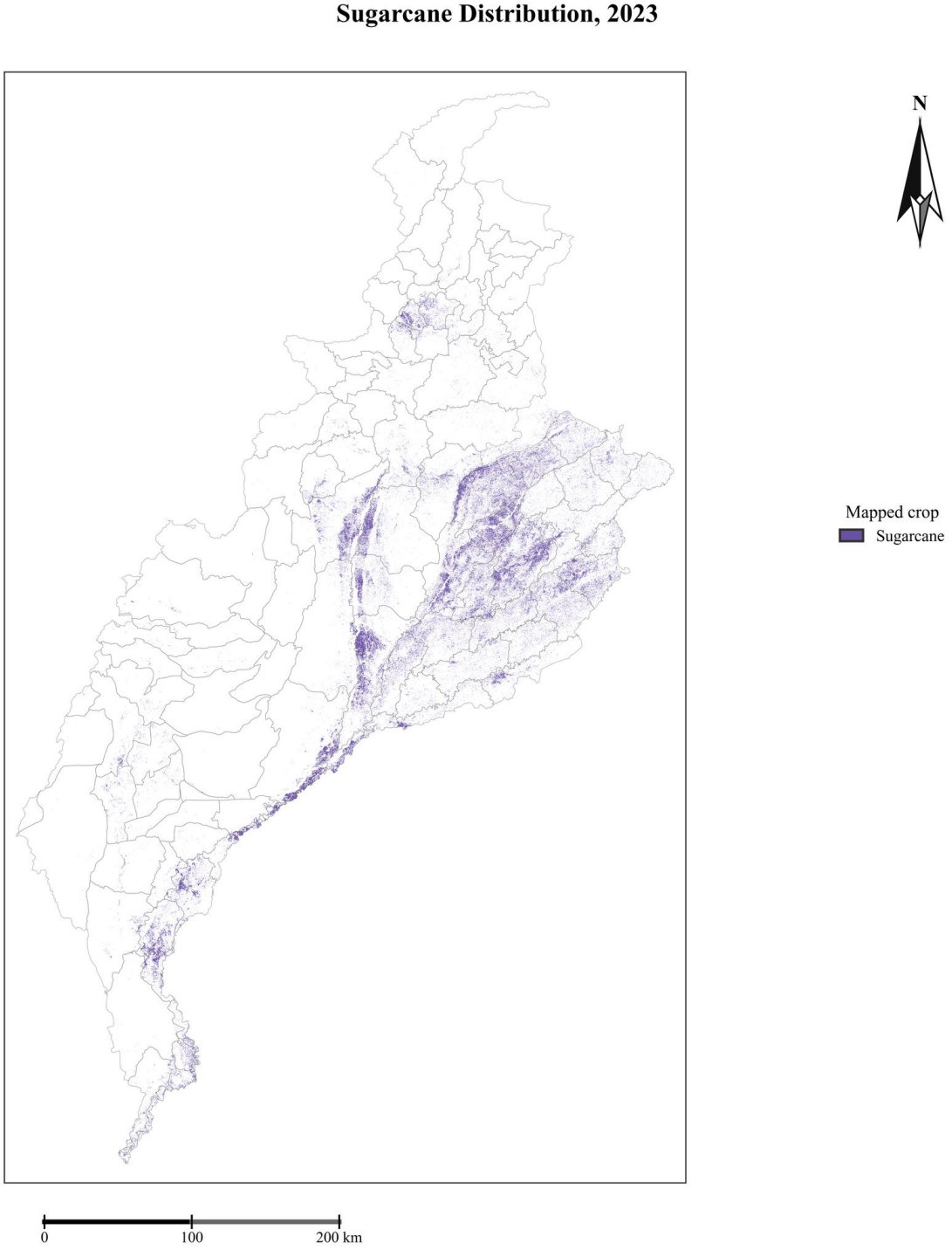


Figure S5. Mapped 2023 sugarcane distribution within the Pakistan portion of the Indus drainage-basin analysis boundary.

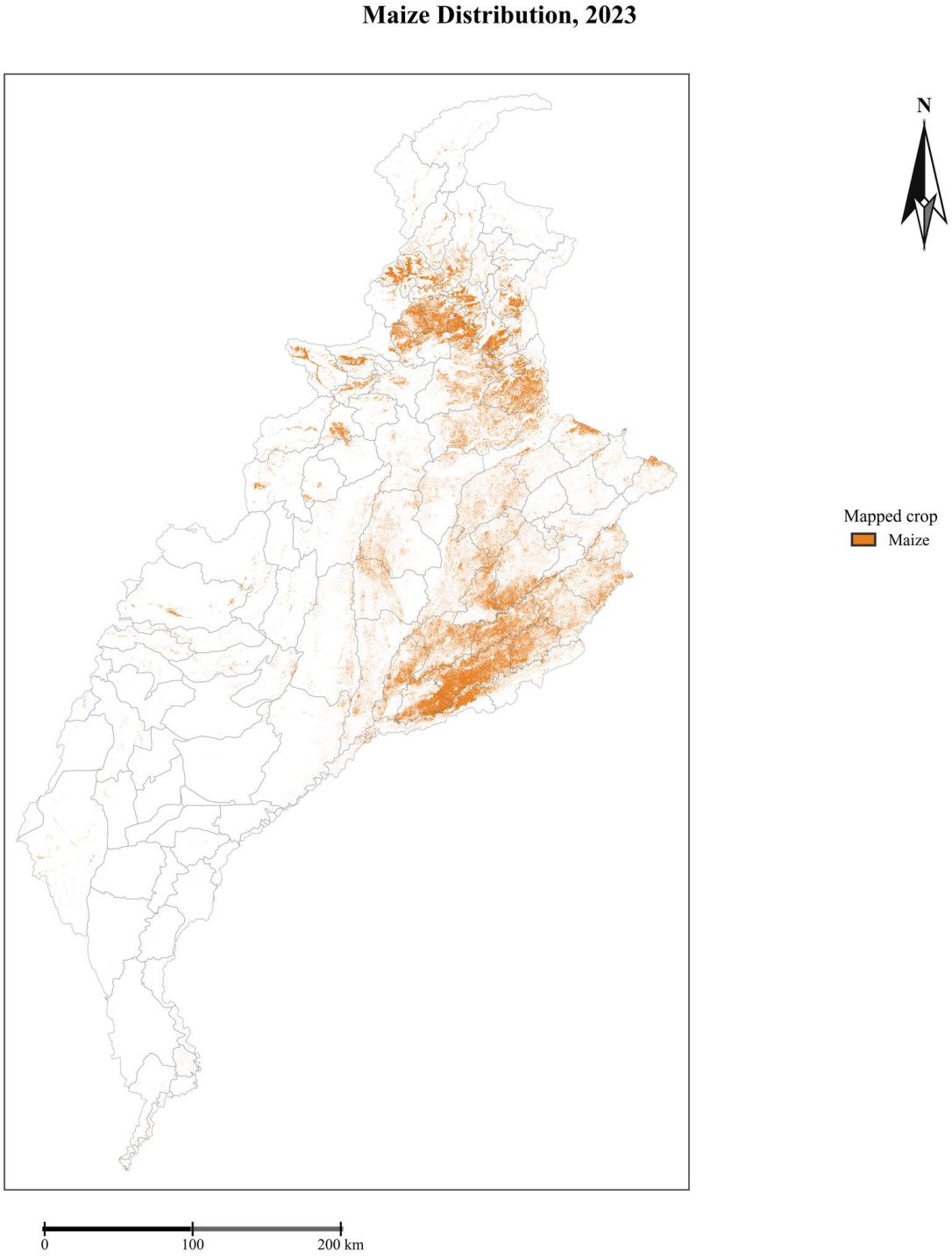


Figure S6. Mapped 2023 maize distribution, combining the spring and autumn crop cycles used in the analysis.

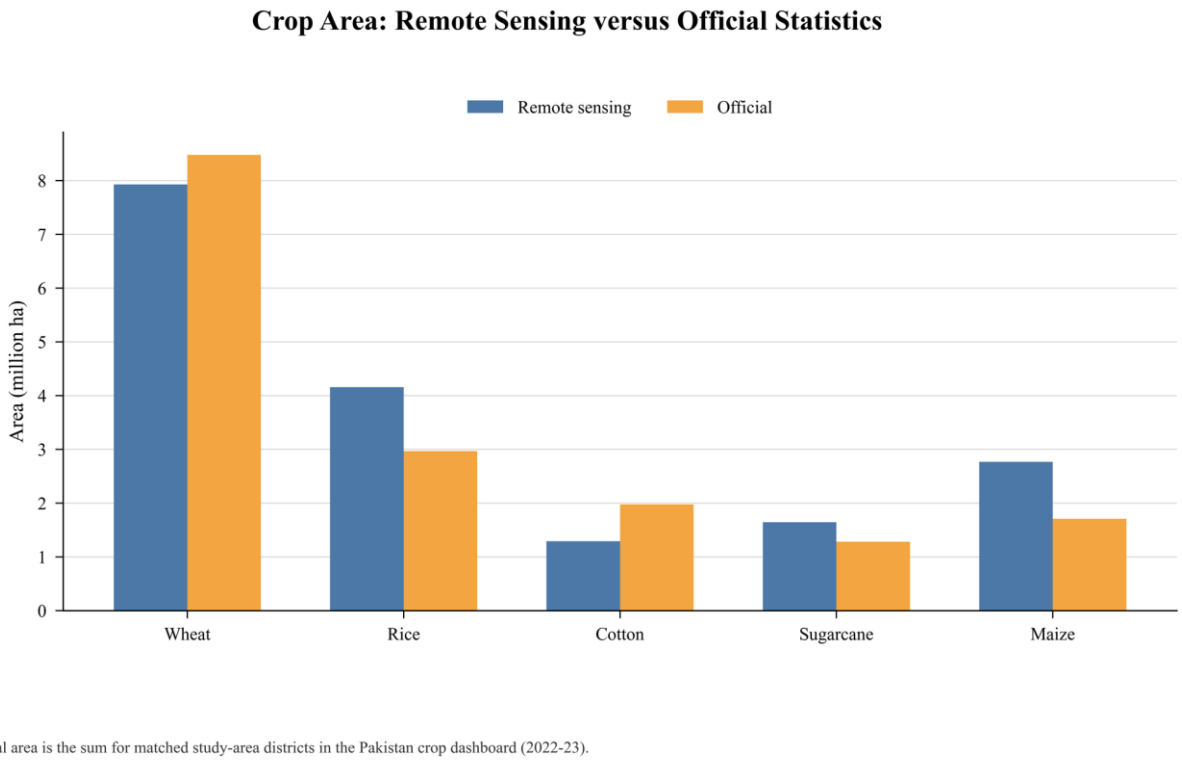


Figure S7. Remote-sensing crop-season area compared with matched official harvested-area statistics for the five target crops.

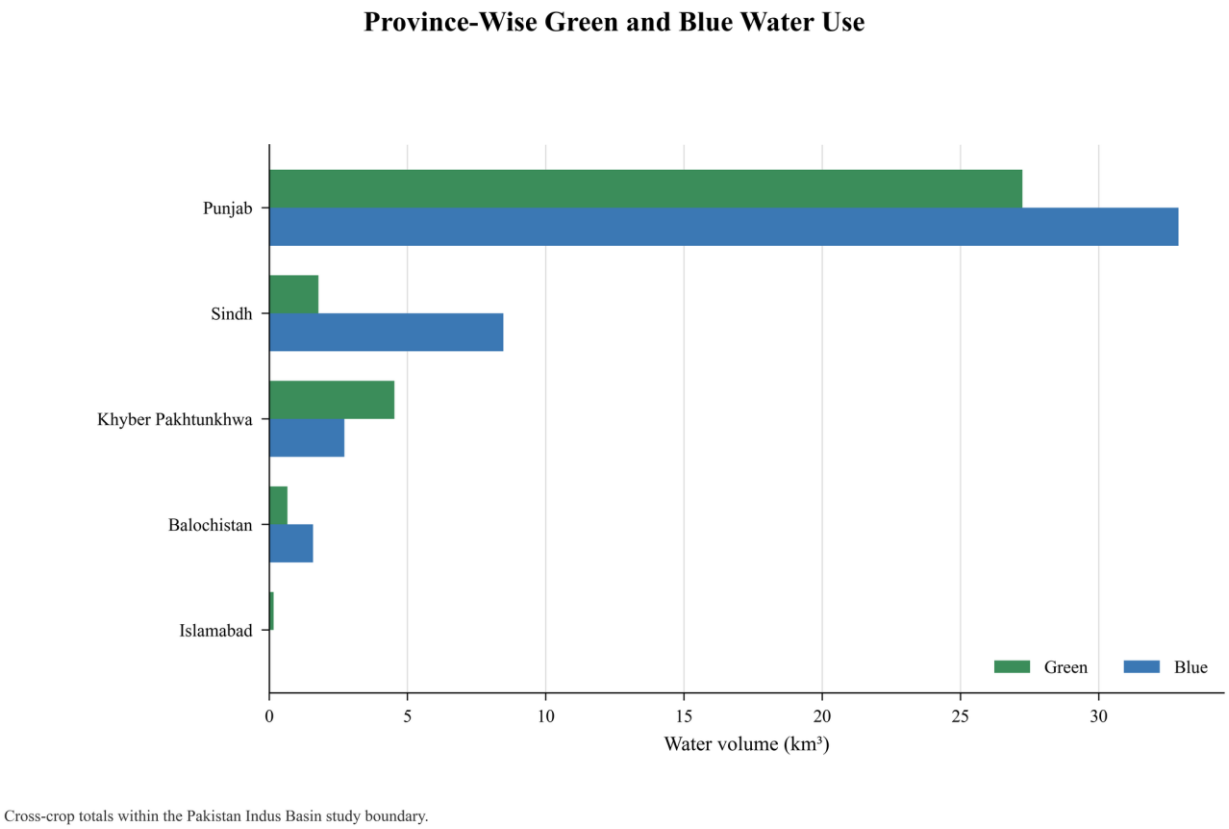


Figure S8. Province-level comparison of rainfall-attributed green and residual-blue consumptive-water volumes.

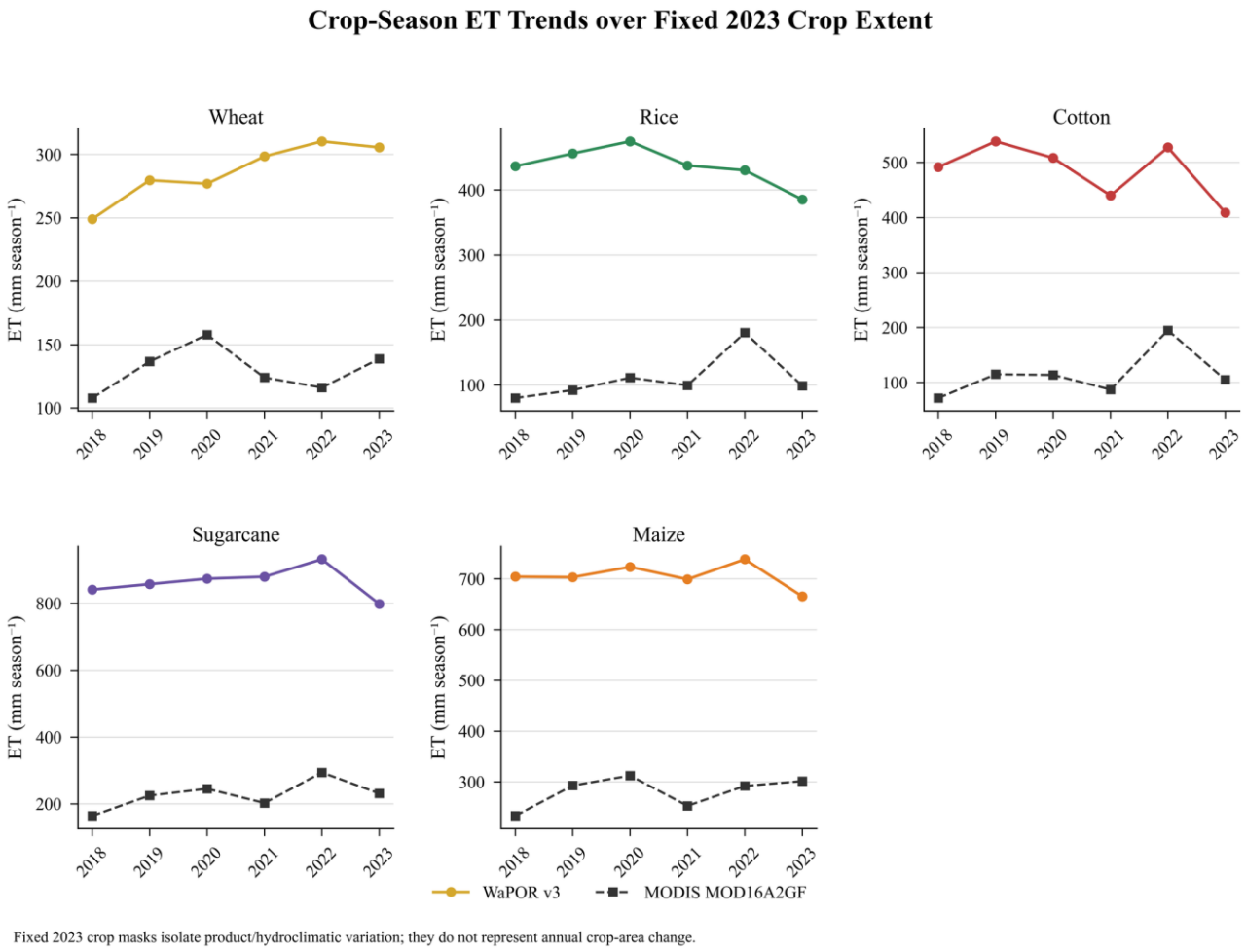


Figure S9. Fixed-2023-crop-extent evapotranspiration context for 2018-2023; the series does not represent historical crop-area change.

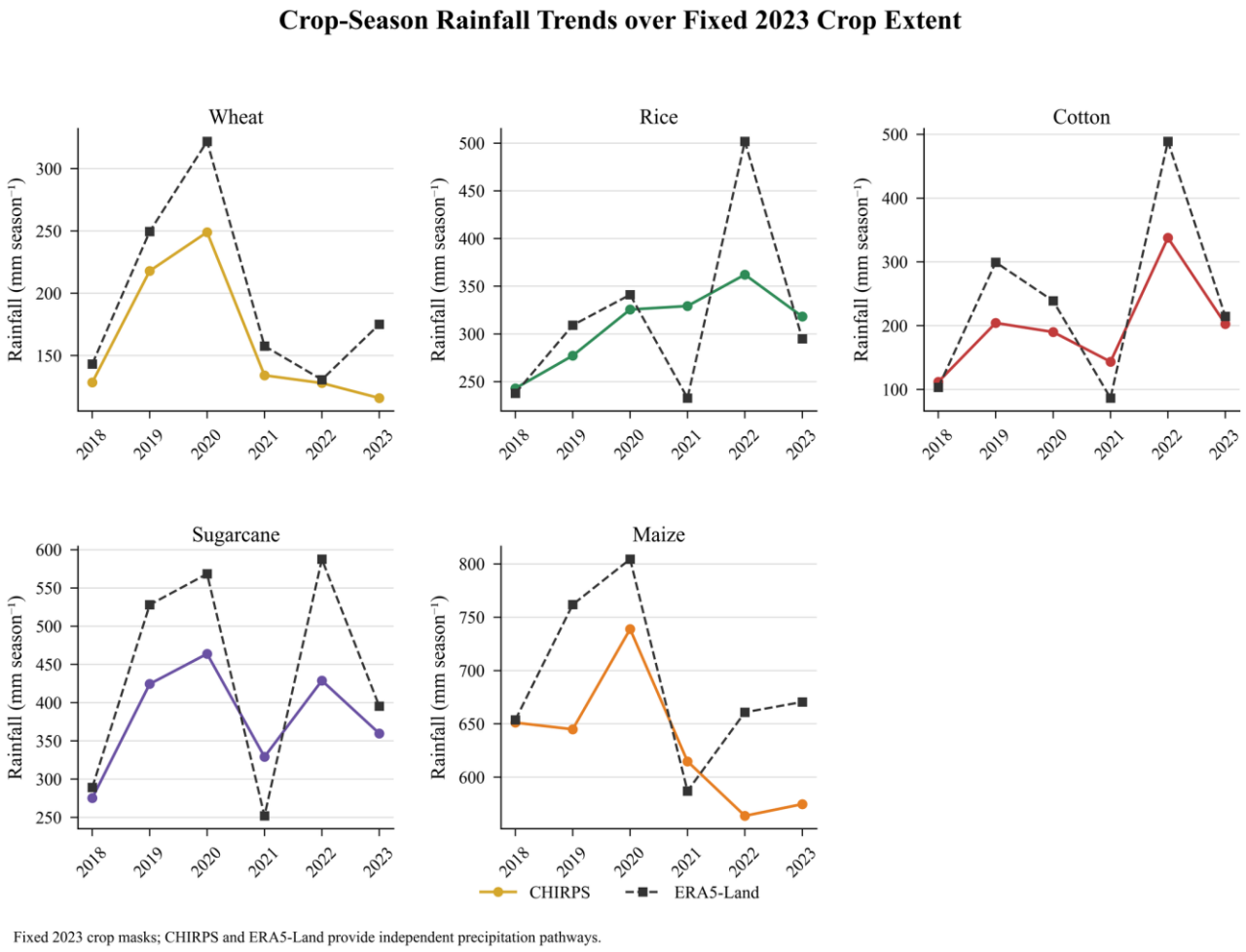


Figure S10. Fixed-2023-crop-extent rainfall context for 2018-2023; the series is a short hydroclimatic diagnostic rather than a climatology.

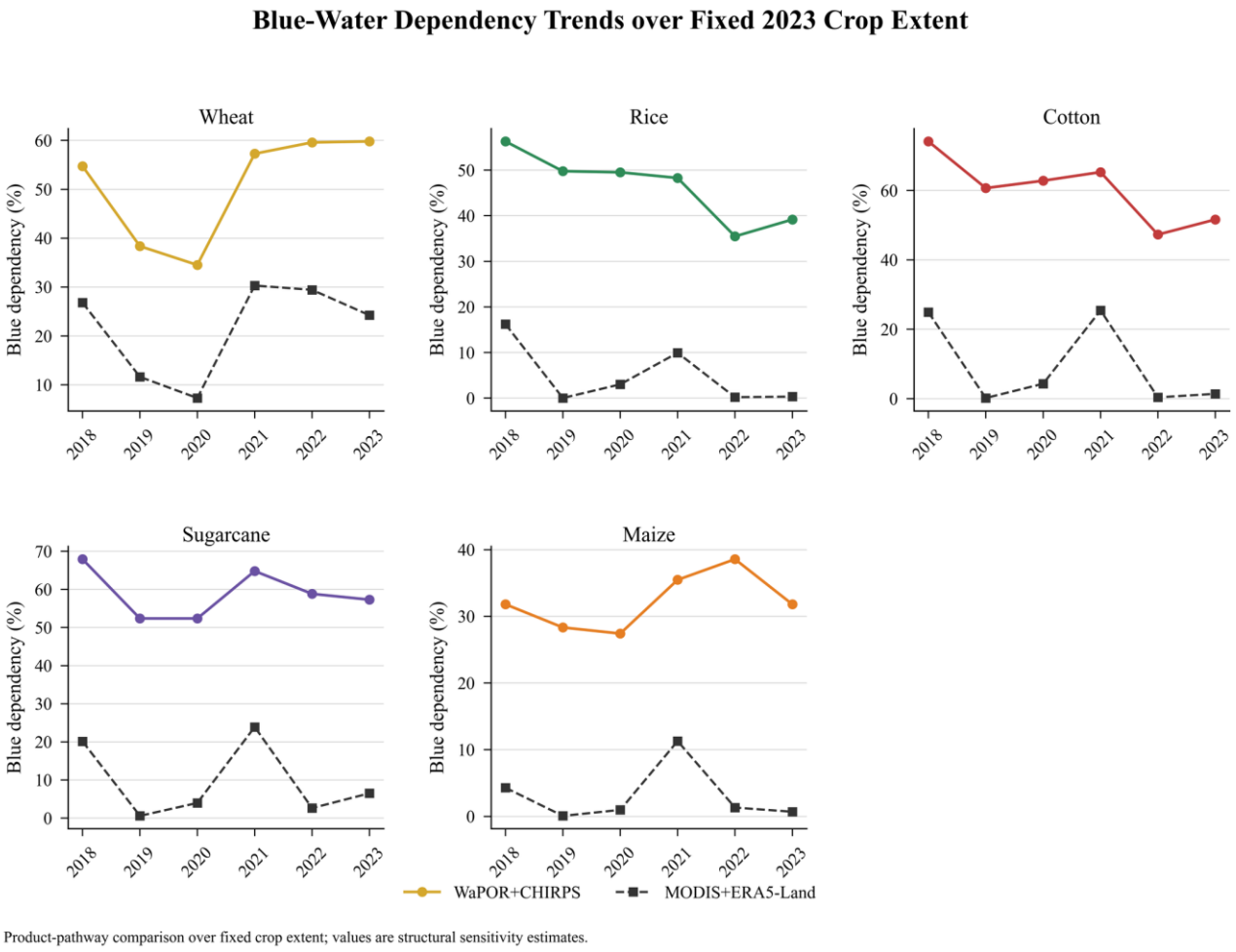


Figure S11. Fixed-2023-crop-extent residual-blue dependency context for 2018-2023.

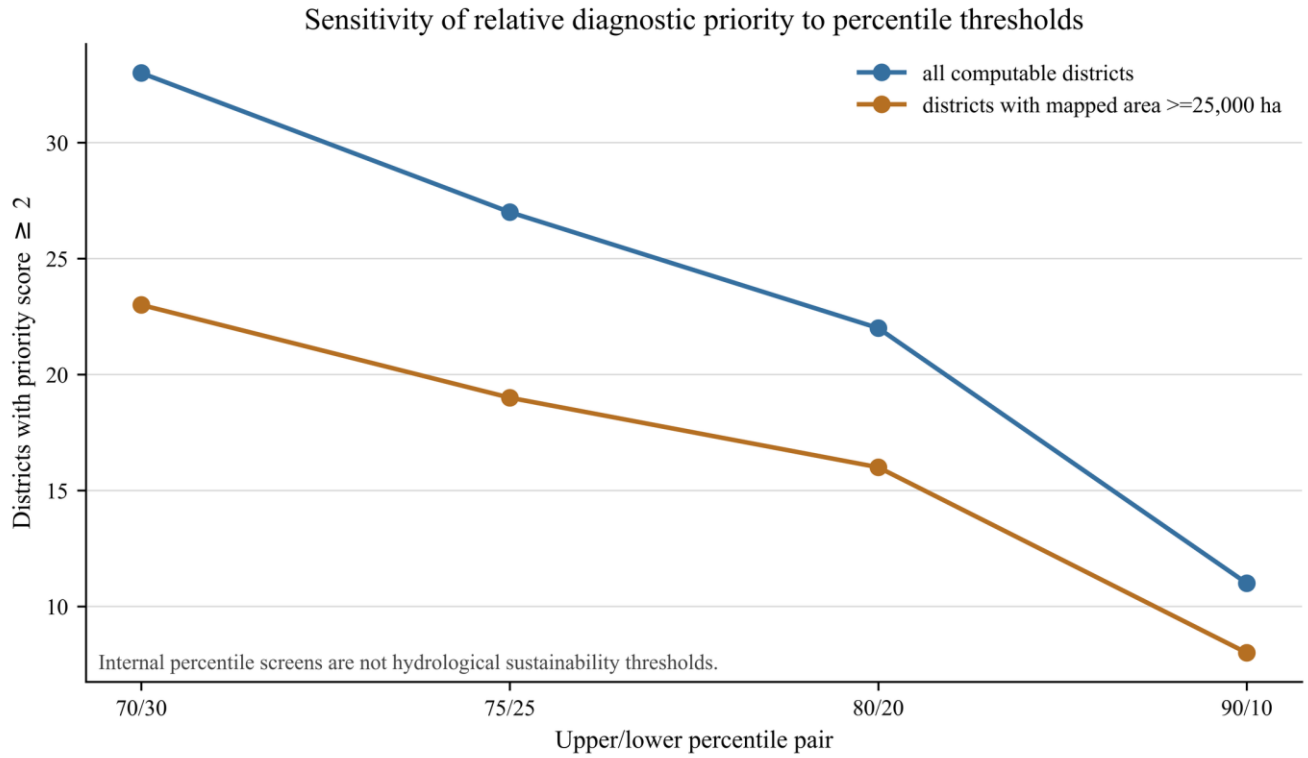


Figure S12. Sensitivity of district hotspot counts to alternative upper- and lower-quantile screening thresholds.

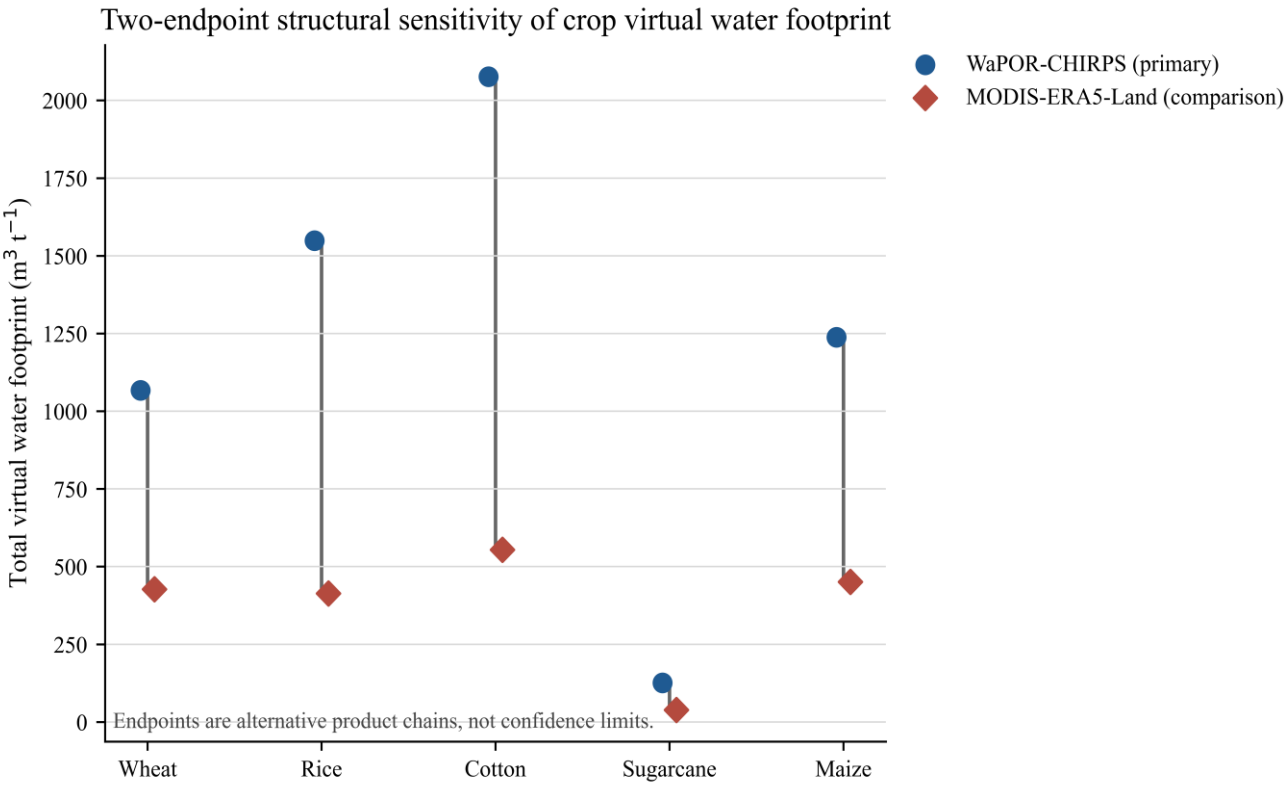
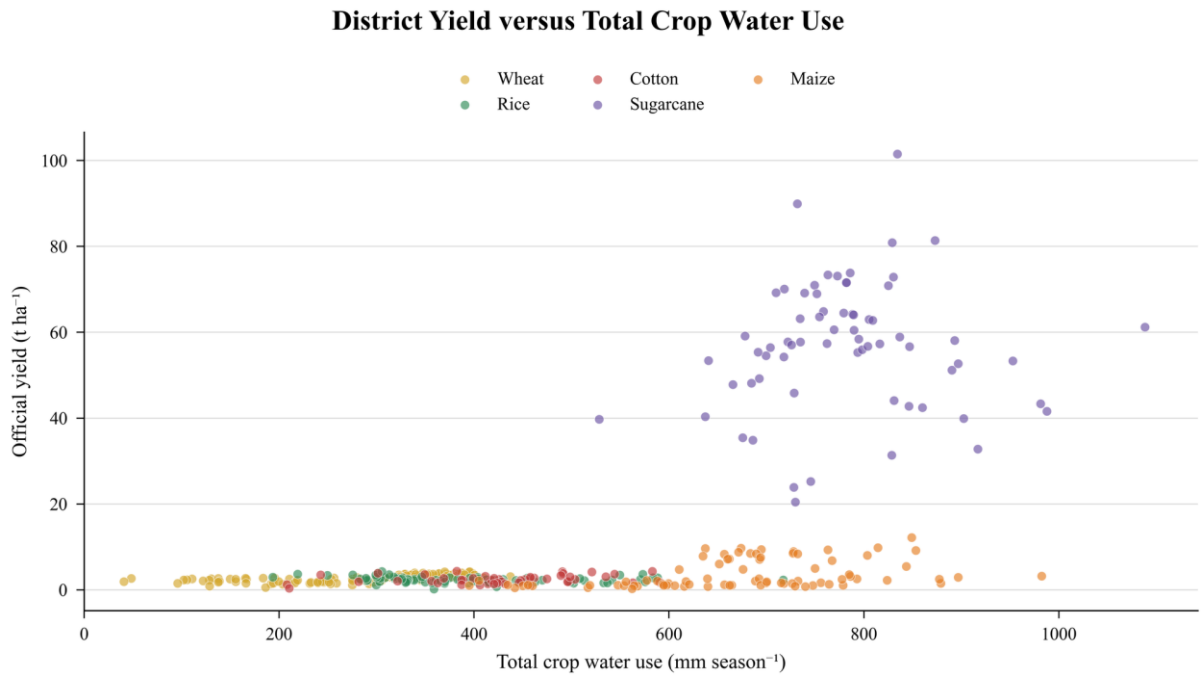
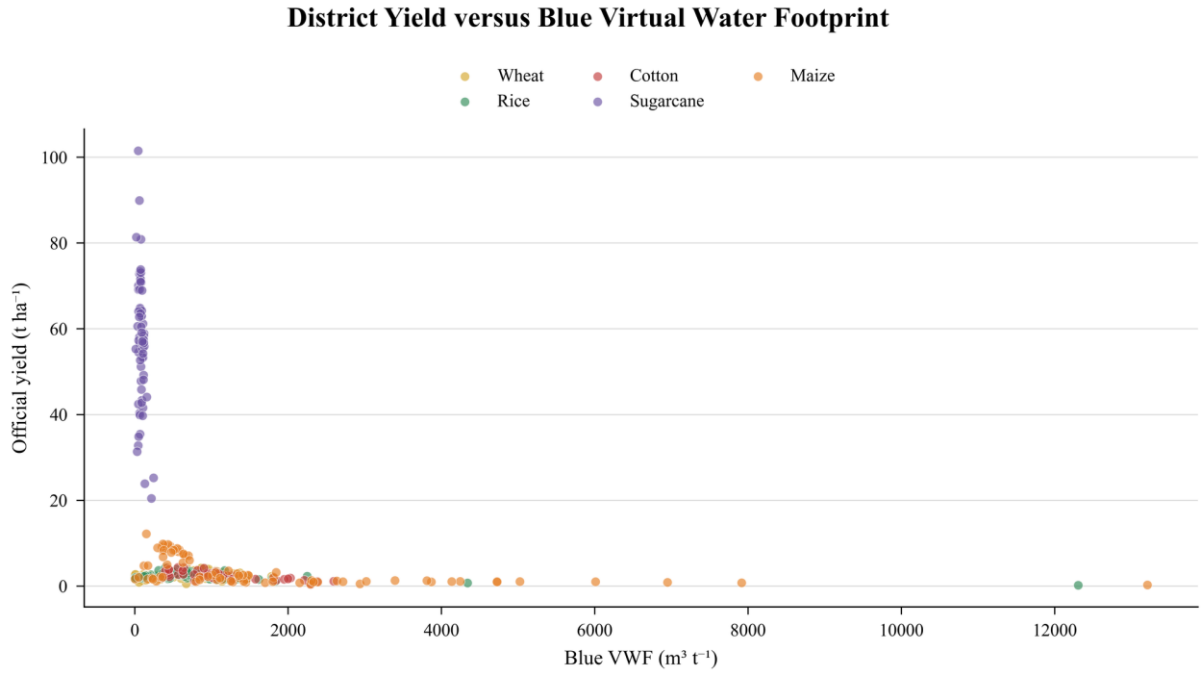


Figure S13. Direct two-endpoint structural sensitivity of crop total virtual water footprint. Circles show the primary WaPOR-CHIRPS estimates, diamonds show the MODIS-ERA5-Land comparison estimates, and connecting lines show the difference between the two evaluated product chains. The endpoints are not confidence limits, and no interpolation or probability distribution was used.



Each point is a district-crop record with positive official yield and mapped crop area.

Figure S14. Relationship between matched crop yield and crop-season consumptive-water use for computable records.



Each point is a district-crop record; blue VWF = blue-water volume divided by mapped production.

Figure S15. Relationship between matched crop yield and residual-blue virtual water footprint for computable records.

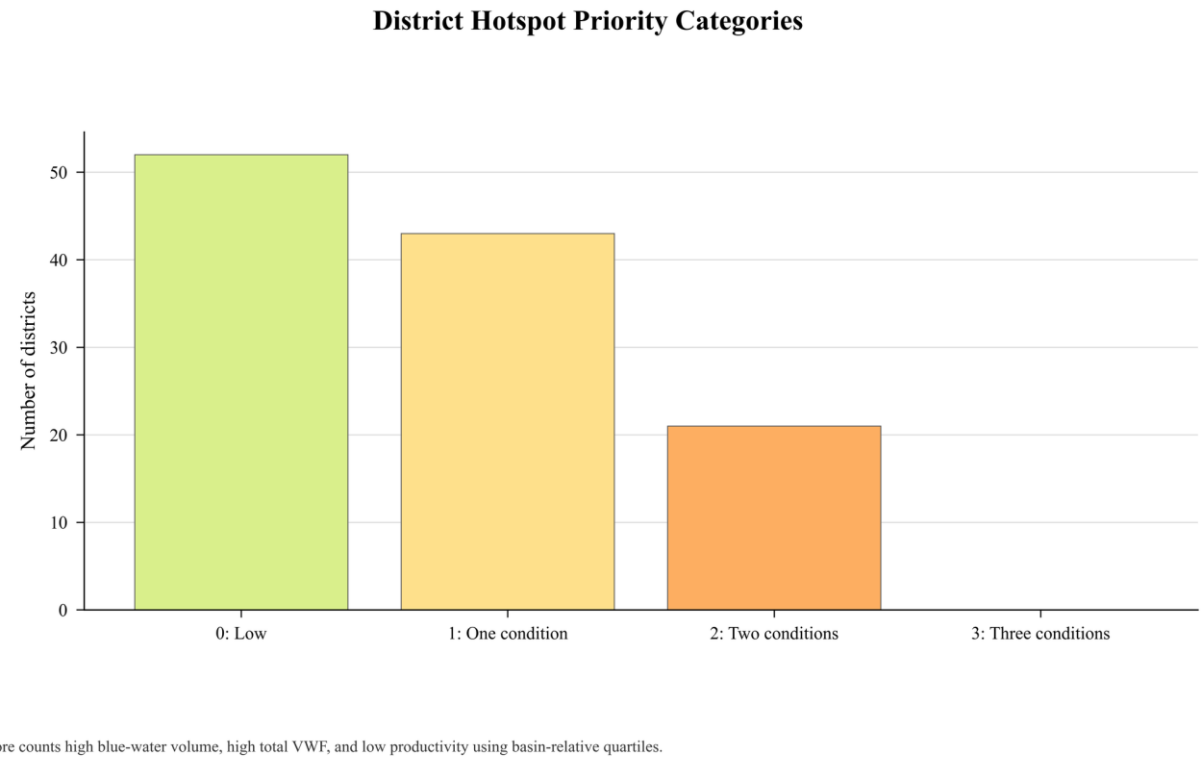


Figure S16. Counts of district records assigned to the exploratory hotspot-priority categories.

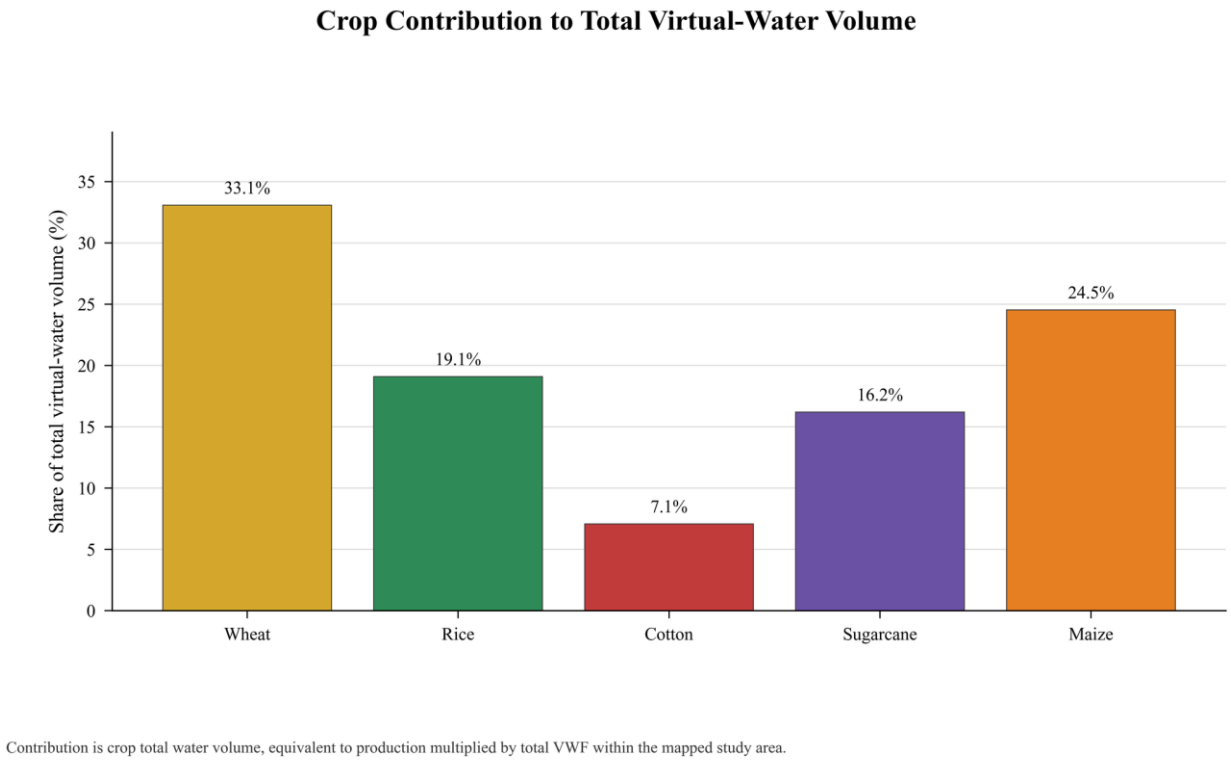


Figure S17. Crop contributions to total virtual-water volume within the mapped 2023 crop-season extent.